

Programming Assignment

Estimating π Using Monte Carlo Simulation

Course: Probability and Mathematical Statistics

1 Problem Background

Consider a square region defined by $|x| \leq 1$ and $|y| \leq 1$ (Total Area = 4) and an inscribed unit circle $x^2 + y^2 \leq 1$ (Area = π).

When random points are uniformly sampled within the square, the probability P of a point falling inside the circle is:

$$P = \frac{\text{Area of Circle}}{\text{Area of Square}} = \frac{\pi}{4}$$

By the **Law of Large Numbers**, π can be estimated via relative frequency:

$$\pi \approx 4 \times \frac{\text{Number of points inside the circle}}{N}$$

2 Assignment Tasks

1. **Simulation Function:** Write a function taking sample size N as input, generating N uniform random points in $[-1, 1] \times [-1, 1]$, and computing the estimated π .
2. **Error Analysis & Curve:** Test a sequence of sample sizes ($N = 10^2$ to 10^6). For each N , calculate the absolute error $|\hat{\pi} - \pi|$ and **plot the error curve** with a logarithmic scale on the x-axis.